



GIRRAWEEN HIGH SCHOOL
MATHEMATICS EXTENSION 1
HSC ASSESSMENT
TASK 1 2025
(December 2024)

Student Number:_____.

		ME12-1	ME12-2	ME12-3	ME12-4	ME12-5	ME12-6	ME12-7
1-5	/5			√				√
6(a)	/3							√
6(b)(c)	/6			√				√
7	/10			√				√
8	/12			√				√
9	/9			√				√
10	/13	√						√
11	/8			√				√
TOTALS	/66	/13		/50				/66

Year 12 Mathematics Extension 1 outcomes

A student:

- ME12-1** applies techniques involving proof or calculus to model and solve problems
- ME12-2** applies concepts and techniques involving vectors and projectiles to solve problems
- ME12-3** applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations
- ME12-4** uses calculus in the solution of applied problems, including differential equations and volumes of solids of revolution
- ME12-5** applies appropriate statistical processes to present, analyse and interpret data
- ME12-6** chooses and uses appropriate technology to solve problems in a range of contexts
- ME12-7** evaluates and justifies conclusions, communicating a position clearly in appropriate mathematical forms

Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↗

Select the alternative A, B, C or D that best answers the question.

1. A ☐ B. ☐ C. ☐ D. ☐

2. A ☐ B. ☐ C. ☐ D. ☐

3. A ☐ B. ☐ C. ☐ D. ☐

4. A ☐ B. ☐ C. ☐ D. ☐

5. A ☐ B. ☐ C. ☐ D. ☐



GIRRAWEEEN HIGH SCHOOL
HSC TASK 1 2025 (*December 2024*)

MATHEMATICS

EXTENSION 1

Time allowed – 90 Minutes
(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- NESA-approved calculators may be used. Reference sheets will be provided.
- Each question attempted is to be marked clearly Question 6, Question 7 etc. Each question is to be answered on a new page in your answer booklet.
- You may ask for spare answer booklets if you need them.
- For Multiple choice: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.
- For Questions 6-11: All necessary working should be shown in each question. Marks may be deducted for careless or badly arranged work.

Multiple Choice begins on the following page

For Multiple choice Questions 1-5: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.

Question 1

Daniel was solving $\sin 2x = \cos x, 0 \leq x \leq 2\pi$. His working was:

$$\sin 2x = \cos x \quad \text{Line 1}$$

$$2 \sin x \cos x = \cos x \quad \text{Line 2}$$

$$\sin x = \frac{1}{2} \quad \text{Line 3}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \quad \text{Line 4}$$

Daniel has made a mistake on one of the lines. The line he made a mistake on was

- (A) Line 1
- (B) Line 2
- (C) Line 3
- (D) Line 4

Question 2

$2 \sin 2x \cos x$ can be rewritten using products to sums as

- (A) $\sin 3x + \sin x$
- (B) $\cos 3x + \cos x$
- (C) $\sin 3x - \sin x$
- (D) $\cos 3x - \cos x$

Question 3

If $2 \sin \theta - 3 \cos \theta$ is arranged in the form $R \cos(\theta + \alpha)$ then $\sin \alpha =$

- (A) $\frac{2}{\sqrt{13}}$
- (B) $-\frac{2}{\sqrt{13}}$
- (C) $\frac{3}{\sqrt{13}}$
- (D) $-\frac{3}{\sqrt{13}}$

Multiple choice continues on the following page

Multiple choice (continued)

Question 4

$$2 \sin\left(\frac{\pi}{4} - x\right) \cos\left(\frac{\pi}{4} - x\right) =$$

- (A) $\cos 2x$
- (B) $\sin 2x$
- (C) $2 \cos x$
- (D) $2 \sin x$

Question 5

If $t = \tan \frac{x}{2}$ then $\tan x + \sec x =$

- (A) $\frac{2t+1-t^2}{1+t^2}$
- (B) $\frac{1}{t}$
- (C) $\frac{1+t}{1-t}$
- (D) $\frac{1+t^2}{1-t^2}$

Examination continues on the following page

Question 6 (9 marks) (Start solutions on a new page)**Marks**

- (a) Sketch the region in the number plane where the inequalities $x^2 + y^2 \leq 25$ and $4x - 3y > 0$ hold simultaneously.

3

- (b) Use compound angle formulae to find the exact value of

(i) $\sin 130^\circ \cos 70^\circ - \cos 130^\circ \sin 70^\circ$

2

(ii) $\frac{\tan 110^\circ - \tan 80^\circ}{1 + \tan 110^\circ \tan 80^\circ}$

2

- (c) Show that the exact value of $\cos \frac{\pi}{12}$ is $\frac{\sqrt{6} + \sqrt{2}}{4}$.

2**Question 7 (10 Marks) (Start solutions on a new page)**

- (a) (i) Express $\sqrt{3} \cos \theta - \sin \theta$ in the form $R \cos(\theta - \alpha)$ where $R > 0$ and α is in radians.

3

(ii) Hence solve $\sqrt{3} \cos \theta - \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$.

2

- (b) Using the auxiliary angle method and choosing a substitution so that $0 < \alpha < 90^\circ$, solve $3 \sin \theta - 2 \cos \theta = 1$ for $0 \leq \theta \leq 360^\circ$.

5

Examination continues on the following page

Question 8 (12 marks) (Start solutions on a new page) **Marks**

- (a) Using the substitution $t = \tan \frac{\theta}{2}$, solve the equation **5**

$$3 \sin \theta + 4 \cos \theta = 2 \text{ for } 0 \leq \theta \leq 360^\circ.$$

- (b) (i) Express $\sqrt{2} \cos x + \sqrt{2} \sin x$ in the form $R \sin(x + \alpha)$. **4**

- (ii) Hence sketch $y = \sqrt{2} \cos x + \sqrt{2} \sin x$ for $0 \leq x \leq 2\pi$. **3**

Question 9 (9 Marks) (Start solutions on a new page)

- (a) Solve for x , $0 \leq x \leq 360^\circ$: **4**

$$\cos 2x - \sin x = \frac{1}{2}$$

- (b) (i) Show using sums to products formulae that **2**

$$\sin 8\theta + \sin 6\theta - \sin 4\theta - \sin 2\theta = 4 \cos 5\theta \sin 2\theta \cos \theta$$

- (ii) Hence solve for θ , $0 \leq \theta \leq 2\pi$ **3**

$$\sin 8\theta + \sin 6\theta - \sin 4\theta - \sin 2\theta = 0$$

Question 10 (13 marks) (Start solutions on a new page)

- (a) Prove by mathematical induction for all integers $n \geq 1$ **4**

$$1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + n(n+1) = \frac{1}{3}n(n+1)(n+2)$$

- (b) Prove by mathematical induction for all integers $n \geq 1$: **4**

$$5^{2n} - 3^{2n} \text{ is divisible by } 16.$$

- (c) Prove by mathematical induction for all integers $n \geq 1$: **5**

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

Examination continues on the following page

Question 11 (8 marks) (Start solutions on a new page)

Marks

(i) Prove that $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$

3

(ii) Use sums to products formulae to show that

2

$$\cos \frac{4\pi}{9} + \cos \frac{2\pi}{9} = \cos \frac{\pi}{9}$$

(iii) Hence use (i) and (ii) to show that $\cos \frac{\pi}{9}$ is a solution to the equation $8x^3 - 6x - 1 = 0$

3

END OF EXAMINATION

Mathematics Assessment Solutions

Girraween High School

Page 1

☐ Year 11

☒ Extension 1

Year: 2025

☐ Year 12

☐ Extension 2

Task Number 1

Suggested solutions

Marker's Feedback

Multiple Choice:

(1) (C) Line 3 was where the error was.

Dividing by $\cos x$ eliminated solutions where $\cos x = 0 \rightarrow$ namely $x = \frac{\pi}{2}$ & $x = \frac{3\pi}{2}$.

(2) (A) As $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$

$$2 \sin x \cos x = \sin(2x+x) + \sin(2x-x) \\ = \sin 3x + \sin x.$$

(3) $2 \sin \theta - 3 \cos \theta = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$

(B) Equating parts

$$2 \sin \theta = -R \sin \theta \sin \alpha \quad | \quad -3 \cos \theta = R \cos \theta \cos \alpha \\ -2 = R \sin \alpha \quad (1) \quad -3 = R \cos \alpha \quad (2)$$

$$(1)^2 + (2)^2: R^2 (\cos^2 \alpha + \sin^2 \alpha) = 2^2 + 3^2$$

$$R = \sqrt{13}.$$

$$\text{Sub. } R = \sqrt{13} \text{ in (1)} \quad -2 = \sqrt{13} \sin \alpha$$

$$\frac{-2}{\sqrt{13}} = \sin \alpha.$$

(4) $2 \sin\left(\frac{\pi}{4} - x\right) \cos\left(\frac{\pi}{4} - x\right)$

$$(A) = \sin\left(\frac{\pi}{2} - 2x\right) \\ = \cos 2x$$

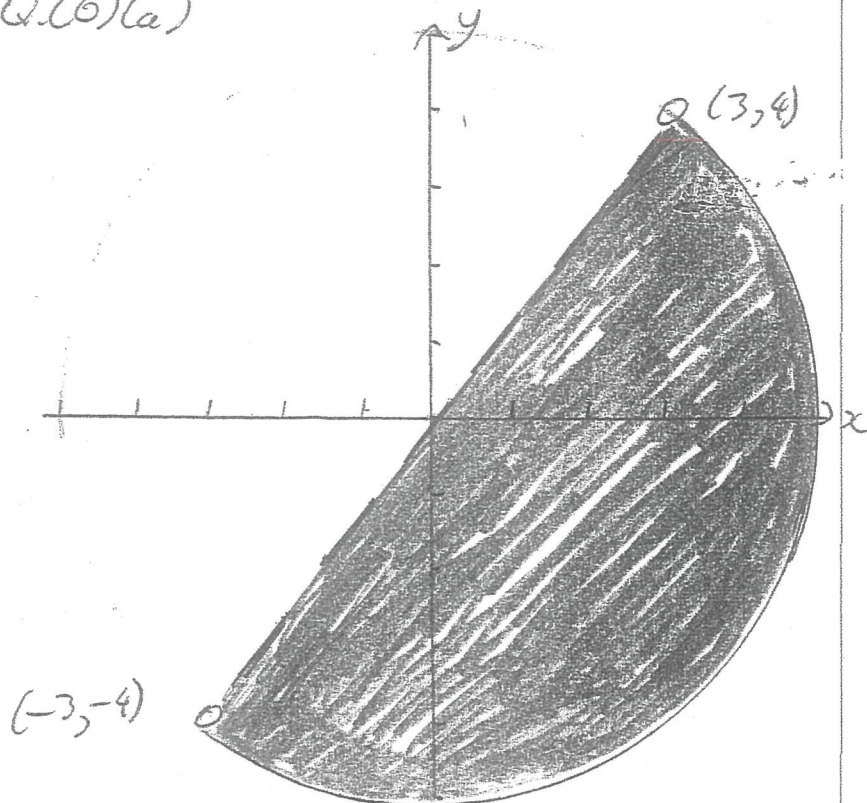
(5) $\tan x + \sec x$

$$(C) = \frac{2t}{1-t^2} + \frac{1+t^2}{1-t^2} \\ = \frac{(1+t)^2}{(1-t)(1+t)} \\ = \frac{1+t}{1-t}.$$

Suggested solutions

Marker's Feedback

Q.6(a)

Note: $4x - 3y > 0$

$$4x > 3y$$

$$\frac{4}{3} > y \text{ or } y < \frac{4}{3}x$$

$$(b)(i) \sin 130^\circ \cos 70^\circ - \cos 130^\circ \sin 70^\circ$$

$$= \sin(130^\circ - 70^\circ)$$

$$= \sin 60^\circ$$

$$= \frac{\sqrt{3}}{2}$$

$$(ii) \frac{\tan 110^\circ - \tan 80^\circ}{1 + \tan 110^\circ \tan 80^\circ}$$

$$= \tan(110^\circ - 80^\circ)$$

$$= \tan 30^\circ$$

$$= \frac{1}{\sqrt{3}}$$

$$(c) \cos \frac{\pi}{12} = \cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$$

$$= \cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6}$$

$$= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4} = \text{RHS.}$$

1 - open circles

1 - solid, dotted lines

1 - correct region

Very few students are required to revise regions otherwise mostly done well. Some students forgot open circles and some dotted & solid lines.

1 - $\sin(130^\circ - 70^\circ)$ 1 - $\frac{\sqrt{3}}{2}$

Mostly done well!

1 - $\tan(110^\circ - 80^\circ)$ 1 - $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{2}$

Mostly done well.

1 - mark for $\cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$

1 - $\frac{\sqrt{6} + \sqrt{2}}{4}$ with $\frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2}$ as a preceding step.

Any other method used then student got a mark however the second mark was only

given if they could arrive to $\frac{\sqrt{6} + \sqrt{2}}{4}$ with correct working out.

Suggested solutions	Marker's Feedback
Q. (7)(a)(i) $\sqrt{3}\cos\theta - \sin\theta \equiv R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$. (ii) $R > 0$ Equating parts, $\sqrt{3}\cos\theta = R\cos\theta\cos\alpha \quad \quad -\sin\theta = R\sin\theta\sin\alpha$ $\sqrt{3} = R\cos\alpha(1) \quad \quad -1 = R\sin\alpha(2)$ $(1)^2 + (2)^2:$ $R^2(\cos^2\alpha + \sin^2\alpha) = (\sqrt{3})^2 + (-1)^2$ $R^2 = 4 \quad R = 2$ Sub. $R=2$ in (1): $\sqrt{3} = 2\cos\alpha$ $\frac{\sqrt{3}}{2} = \cos\alpha$ $\alpha = \frac{11\pi}{6}$ $\therefore \sqrt{3}\cos\theta - \sin\theta \equiv 2\cos(\theta - \frac{11\pi}{6})$ (ii) Hence $\sqrt{3}\cos\theta - \sin\theta = 1$. $2\cos(\theta - \frac{11\pi}{6}) = 1$ $\cos(\theta - \frac{11\pi}{6}) = \frac{1}{2}$ Noting $-\frac{11\pi}{6} \leq \theta - \frac{11\pi}{6} \leq \frac{\pi}{6}$. $\theta - \frac{11\pi}{6} = -\frac{5\pi}{3}, -\frac{\pi}{3}$ $\theta = \frac{\pi}{6}, \frac{3\pi}{2}$	Some students couldn't find the correct value for α Common errors: - $\alpha = \frac{\pi}{6}, \alpha = \frac{5\pi}{6}$ * Mark deducted if they didn't justify α in 4th quadrant * For part (i), if they got wrong value for α and as a result the domain change made part (ii) question easier, then no carry forward mark given for finding θ. * Some students didn't pay attention to domain when finding values of θ. * Mark deducted if θ not given in radians.

Suggested solutions	Marker's Feedback
Q. (7)(b) $3\sin\theta - 2\cos\theta = R\sin(\theta - \alpha), R > 0$ $3\sin\theta - 2\cos\theta = R\sin\theta\cos\alpha - R\cos\theta\sin\alpha$ Equating parts $3\sin\theta = R\sin\theta\cos\alpha \quad \quad -2\cos\theta = -R\cos\theta\sin\alpha$ $3 = R\cos\alpha(1) \quad \quad 2 = R\sin\alpha(2)$ $(1)^2 + (2)^2: R^2(\cos^2\alpha + \sin^2\alpha) = 3^2 + 2^2$ $R = \sqrt{13}$ Sub. $R = \sqrt{13}$ in (1): Sub. $R = \sqrt{13}$ in (2): $3 = \sqrt{13}\cos\alpha \quad \quad 2 = \sqrt{13}\sin\alpha$ $\frac{3}{\sqrt{13}} = \cos\alpha \quad \quad \frac{2}{\sqrt{13}} = \sin\alpha$ $\alpha = 33^\circ 41'!$ 1 for $\underline{5}$ Hence solving $3\sin\theta - 2\cos\theta = 1$ $\sqrt{13}\sin(\theta - 33^\circ 41') = 1$ $\sin(\theta - 33^\circ 41') = \frac{1}{\sqrt{13}}$ $\theta - 33^\circ 41' = 16^\circ 6', 163^\circ 54'!$ $\theta = 49^\circ 47' \quad \quad , 197^\circ 35'!$	for right pick * Some students didn't choose the right form $R\sin(\theta - \alpha)$ * Need to pay attention to $0 < \alpha < 90^\circ$ * A few students didn't list all possible solutions. * Mark deducted when θ not given in degrees.

Suggested solutions	Marker's Feedback
Q.18)(a) $3\sin\theta + 4\cos\theta = 2$ $\frac{3 \times 2t}{1+t^2} + \frac{4(1-t^2)}{1+t^2} = 2.$ $6t + 4 - 4t^2 = 2 + 2t^2$ $0 = 6t^2 - 6t - 2$ $0 = 3t^2 - 3t - 1.$ $t = \frac{3 \pm \sqrt{(-3)^2 - 4 \times 3 \times -1}}{2 \times 3}$ $= \frac{3 \pm \sqrt{21}}{6}.$ $\tan\left(\frac{\theta}{2}\right) = \frac{3 + \sqrt{21}}{6} \quad \left \quad \tan\left(\frac{\theta}{2}\right) = \frac{3 - \sqrt{21}}{6}\right.$ $\frac{\theta}{2} = 51^\circ 39' \quad \left \quad \frac{\theta}{2} = 165^\circ 13'\right.$ $\theta = 103^\circ 18' \quad \left \quad \theta = 330^\circ 27'\right.$ Test: $\theta = 180^\circ$ $3\sin 180^\circ + 4\cos 180^\circ = -4 \neq 2.$ 180° is not a solution $\therefore$ Solutions are $\theta = \underline{103^\circ 18', 330^\circ 27'}.$ 5	Lots of wrong answers for the solution of quadratic. Many students have difficulty in finding the correct solution for $\tan \frac{\theta}{2} = \frac{3 - \sqrt{21}}{6}$ Some forgot to Test for 180°

Suggested solutions	Marker's Feedback
Q. (8)(b)(i) $\sqrt{2} \cos x + \sqrt{2} \sin x \equiv R \sin(x + \alpha)$ $\sqrt{2} \cos x + \sqrt{2} \sin x \equiv R \sin x \cos \alpha + R \cos x \sin \alpha$ Equating parts, $\sqrt{2} \cos x = R \cos x \sin \alpha$ $\sqrt{2} \sin x = R \sin x \cos \alpha$ $\sqrt{2} = R \sin \alpha (1)$ $\sqrt{2} = R \cos \alpha (2)$ $(1)^2 + (2)^2: R^2 (\cos^2 \alpha + \sin^2 \alpha) = (\sqrt{2})^2 + (\sqrt{2})^2$ $R^2 = 4$ $R = 2$ Sub. $R = 2$ in (1): $\sqrt{2} = 2 \sin \alpha$ Sub. $R = 2$ in (2): $\frac{\sqrt{2}}{2} = \sin \alpha$ $\frac{\sqrt{2}}{2} = \cos \alpha$ $\alpha = \frac{\pi}{4}$	Some wrote $\tan \alpha = 1$ $\alpha = \frac{\pi}{4}$ They lost 1 mark.
$\therefore \sqrt{2} \cos x + \sqrt{2} \sin x = 2 \sin(x + \frac{\pi}{4})$ (ii) $\rightarrow$ Period = 2π. Amplitude = 2. Phase = $\frac{\pi}{4}$ $x = 0$ (or 2π) $y = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$	3
	Common mistakes * wrong y intercept * not writing all the x values ie $0, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, 2\pi$

Suggested solutions	Marker's Feedback
Q.9)(a) $\cos 2x - \sin x = \frac{1}{2}$ $(1 - 2\sin^2 x) - \sin x = \frac{1}{2}$ $0 = 4\sin^2 x + 2\sin x - 1$ $\sin x = \frac{-2 \pm \sqrt{(-2)^2 - 4 \times 4 \times -1}}{2 \times 4}$ $= \frac{-2 \pm 2\sqrt{5}}{8}$ $= \frac{-1 \pm \sqrt{5}}{4}$ 4 If $\sin x = \frac{-1 + \sqrt{5}}{4}$ 1 If $\sin x = \frac{-1 - \sqrt{5}}{4}$ $x = 18^\circ, 162^\circ$ 1 $x = 234^\circ, 306^\circ$ 1	* The vast majority of students knew to use $\cos 2x = 1 - 2\sin^2 x$ & get a quadratic equation using $\sin x$. * A very worrying group of about 10 students forgot the NULL FACTOR law when solving the equation. * Many students didn't realise $-\frac{1+\sqrt{5}}{4}$ and $-\frac{1-\sqrt{5}}{4}$ are
(9)(b) (i) Either $\sin 8\theta + \sin 6\theta - \sin 4\theta - \sin 2\theta$ $= [\sin(7\theta + \theta) + \sin(7\theta - \theta)] - [\sin(3\theta + \theta) + \sin(3\theta - \theta)]$ $= 2\sin 7\theta \cos \theta - 2\sin 3\theta \cos \theta$ $= 2\cos \theta [\sin 7\theta - \sin 3\theta]$ $= 2\cos \theta [\sin(5\theta + 2\theta) - \sin(5\theta - 2\theta)]$ $= 2\cos \theta \times 2\cos 5\theta \sin 2\theta$ 1 2 $= 4\cos 5\theta \sin 2\theta \cos \theta$ $= \text{RHS QED.}$	(a) NOT EQUAL and (b) NOT OPPOSITES so that there are actually 2 sets of 2 solutions $[\sin^{-1}(\frac{-1+\sqrt{5}}{4}) = 18^\circ, 162^\circ$ & $\sin^{-1}(\frac{-1-\sqrt{5}}{4}) = 234^\circ, 306^\circ$
or $\sin 8\theta - \sin 4\theta + \sin 6\theta - \sin 2\theta$ $= \sin(6\theta + 2\theta) - \sin(6\theta - 2\theta) + \sin(4\theta + 2\theta) - \sin(4\theta - 2\theta)$ $= 2\cos 6\theta \sin 2\theta + 2\cos 4\theta \sin 2\theta$ $= 2\sin 2\theta (\cos 6\theta + \cos 4\theta)$ $= 2\sin 2\theta [\cos(5\theta + \theta) + \cos(5\theta - \theta)]$ $= 2\sin 2\theta [2\cos 5\theta \cos \theta]$ 1 $= 4\cos 5\theta \sin 2\theta \cos \theta.$	or $\sin 8\theta - \sin 2\theta + \sin 6\theta - \sin 4\theta$ $= \sin(5\theta + 3\theta) - \sin(5\theta - 3\theta)$ $+ \sin(5\theta + \theta) - \sin(5\theta - \theta)$ $= 2\cos 5\theta \sin 3\theta + 2\cos 5\theta \sin \theta$ 1 $= 2\cos 5\theta [\sin 3\theta + \sin \theta]$ $= 2\cos 5\theta [\sin(2\theta + \theta) + \sin(2\theta - \theta)]$ $= 2\cos 5\theta \times 2\sin 2\theta \cos \theta$ 1 $= 4\cos 5\theta \sin 2\theta \cos \theta.$

(9)(b)(i) was mostly well done.

Suggested solutions.

Marker's Feedback

Q(9)(b)(ii) Here solving

$$\sin 8\theta + \sin 6\theta - \sin 4\theta - \sin 2\theta = 0$$

$$4 \cos 5\theta \sin 2\theta \cos \theta = 0$$

$$0 \leq \theta \leq 2\pi$$

$$\text{Either } \cos 5\theta = 0 \quad 0 \leq 5\theta \leq 10\pi$$

$$5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2},$$

$$\frac{11\pi}{2}, \frac{13\pi}{2}, \frac{15\pi}{2}, \frac{17\pi}{2}, \frac{19\pi}{2}$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{\pi}{2}, \frac{7\pi}{10}, \frac{9\pi}{10},$$

$$\frac{11\pi}{10}, \frac{13\pi}{10}, \frac{3\pi}{2}, \frac{17\pi}{10}, \frac{19\pi}{10}$$

$$0 \leq \theta \leq 2\pi$$

$$\text{or } \sin 2\theta = 0 \quad 0 \leq 5\theta \leq 10\pi$$

$$2\theta = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

Already in $\cos 5\theta = 0$

$$\text{or } \cos \theta = 0$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2} \rightarrow \text{already in } \cos 5\theta = 0.$$

Final solutions:

$$\theta = 0, \frac{\pi}{10}, \frac{3\pi}{10}, \frac{\pi}{2}, \frac{7\pi}{10}, \frac{9\pi}{10}, \pi,$$

$$\frac{11\pi}{10}, \frac{13\pi}{10}, \frac{3\pi}{2}, \frac{17\pi}{10}, \frac{19\pi}{10}, 2\pi.$$

A minority of students had no idea, but most students realised that there were 3 possibilities $\cos 5\theta = 0$, $\sin 2\theta = 0$ or $\cos \theta = 0$.

However, for $\cos 5\theta = 0$ MOST students forgot if $0 \leq \theta \leq 2\pi$ then $0 \leq 5\theta \leq 10\pi$

which meant they missed the solutions

$$\frac{7\pi}{10}, \frac{9\pi}{10}, \dots \text{ to } \frac{19\pi}{10}.$$

1 for $\cos 5\theta = 0$

or $\sin 2\theta = 0$

or $\cos \theta = 0$

1 for getting

some ~~one~~ of the answers

& 1 for ALL answers.

Q.10) (a) Step 1: Show true for $n=1$:

$$\begin{array}{ll} \text{LHS} & \text{RHS} \\ = 1 \times 2 & = \frac{1}{3} \times 1 \times (1+1)(1+2) \\ = 2 & = 2 \end{array}$$

LHS = RHS $\Rightarrow$ True for $n=1$.

Step 2: Assume true for $n=k$

i.e. $1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{1}{3} k(k+1)(k+2)$.

Step 3: Prove true for $n=k+1$

i.e. RTP $1 \times 2 + \dots + k(k+1) + (k+1)(k+2)$
 $= \frac{1}{3} (k+1)(k+2)(k+3)$.

including stat

$$\begin{aligned} & \downarrow \text{LHS} \\ & 1 \times 2 + \dots + k(k+1) + (k+1)(k+2) \quad] \text{ for } \text{assumption} \\ & = \frac{1}{3} k(k+1)(k+2) + (k+1)(k+2) \quad [\text{By assumption}] \\ & = \frac{1}{3} k(k+1)(k+2) + \frac{1}{3} \times 3(k+1)(k+2) \\ & = \frac{1}{3} (k+1)(k+2)(k+3) \\ & = \text{RHS} \quad \text{QED} \quad 1 \text{ for finish. } \underline{4} \end{aligned}$$

If it is true for $n=k$ it is true for $n=k+1$.
 Hence as it is true for $n=1$ it will be true
 for all positive integers n by the principle
 of mathematical induction.

Mostly done well.

(-1) mark if they
 did not mention
 'by assumption'
 in question 10a only.

Q.(10)(b) Step 1: Show true for $n=1$.

$$\begin{aligned} &\text{LHS} \\ &= 5^2 - 3^2 \end{aligned}$$

$$= 16$$

which is divisible by 16.

Step 2: Assume true for $n=k$

$$\text{i.e. } 5^{2k} - 3^{2k} = 16P, P \text{ an integer.}$$

$$\text{or } 3^{2k} = 5^{2k} - 16P.$$

Step 3: Prove true for $n=k+1$.

$$\text{i.e. RTP: } 5^{2k+2} - 3^{2k+2} = 16Q, Q \text{ an integer.}$$

LHS

$$= 5^{2k+2} - 3^{2k+2}$$

$$= 25 \times 5^{2k} - 9 \times (5^{2k} - 16P) \quad \text{[By assumption]}$$

$$= 25 \times 5^{2k} - 9 \times 5^{2k} + 9 \times 16P$$

$$= 16 \times 5^{2k} + 9 \times 16P$$

$$= 16[5^{2k} + 9P]$$

$$= 16Q, Q = 5^{2k} + 9P, \text{ an integer as } k, P \text{ integer.}$$

If it is true for $n=k$ it will be true for $n=k+1$. Hence as it is true for $n=1$ it will be true for all positive integers n by the principle of mathematical induction.

Mostly done well.

4

Q.10)(c) Step 1: Show true for $n=1$.

$$\begin{array}{ll} \text{LHS} & \text{RHS} \\ = \frac{1}{1 \times 2 \times 3} & = \frac{1(1+3)}{4 \times (1+1)(1+2)} \\ = \frac{1}{6} & = \frac{4}{24} \\ & = \frac{1}{6} \end{array}$$

LHS = RHS. True for $n=1$.

Step 2: Assume true for $n=k$

i.e. $\frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$

Step 3: Prove true for $n=k+1$

i.e. RTP

$$\frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} = \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

$$= \frac{1}{1 \times 2 \times 3} + \dots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \quad \text{(By assumption)}$$

$$= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)}$$

$$= \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)}$$

$$\begin{array}{l} \frac{k^3 + k^2 + 5k^2 + 5k + 4k + 4}{4(k+1)(k+2)(k+3)} \\ = \frac{k^2(k+1) + 5k(k+1) + 4(k+1)}{4(k+1)(k+2)(k+3)} \end{array}$$

NOTE: MUST SHOW HOW from assumption to finish

also use factor theorem & polynomial division here.

$$\begin{array}{l} = \frac{(k+1)(k^2 + 5k + 4)}{4(k+1)(k+2)(k+3)} \\ = \frac{(k+1)(k+4)}{4(k+2)(k+3)} \\ = \text{RHS QED.} \end{array}$$

If it is true for $n=k$, it will be true for $n=k+1$. Hence as it is true for $n=1$ it will be true for all positive integers n by mathematical induction.

Majority got to this step.

Then some students wrote the final step without any explanation and lost one mark. Others who did not finish only got 3.

$$Q.(ii) (i) \cos 4\theta$$

$$= \cos 2(2\theta)$$

$$= 2\cos^2 2\theta - 1$$

$$= 2(2\cos^2 \theta - 1)^2 - 1$$

$$= 2(4\cos^4 \theta - 4\cos^2 \theta + 1) - 1$$

$$= 8\cos^4 \theta - 8\cos^2 \theta + 2 - 1$$

$$= 8\cos^4 \theta - 8\cos^2 \theta + 1$$

$$= \text{RHS QED.}$$

$$(ii) \cos \frac{4\pi}{9} + \cos \frac{2\pi}{9}$$

$$A = \cos\left(\frac{\pi}{3} + \frac{\pi}{9}\right) + \cos\left(\frac{\pi}{3} - \frac{\pi}{9}\right)$$

$$= 2\cos \frac{\pi}{3} \cos \frac{\pi}{9}$$

$$[B \cos(A+B) + \cos(A-B) = 2\cos A \cos B]$$

$$= 2 \times \frac{1}{2} \times \cos \frac{\pi}{9}$$

$$= \cos \frac{\pi}{9}$$

$$= \text{RHS QED.}$$

$$(iii) \text{ Letting } \theta = \frac{\pi}{9}, \text{ this means}$$

$$\cos 4\theta + \cos 2\theta = \cos \theta$$

$$[8\cos^4 \theta - 8\cos^2 \theta + 1] + [2\cos^2 \theta - 1] = \cos \theta$$

$$8\cos^4 \theta - 6\cos^2 \theta = \cos \theta$$

$$8\cos^4 \theta - 6\cos^2 \theta - \cos \theta = 0$$

$$\therefore 8\cos^4 \frac{\pi}{9} - 6\cos^2 \frac{\pi}{9} - \cos \frac{\pi}{9} = 0$$

$$\cos \frac{\pi}{9} [8\cos^3 \frac{\pi}{9} - 6\cos \frac{\pi}{9} - 1] = 0$$

$$\text{As } \cos \frac{\pi}{9} \neq 0,$$

$$8\cos^3 \frac{\pi}{9} - 6\cos \frac{\pi}{9} - 1 = 0$$

$$\therefore \cos \frac{\pi}{9} \text{ is a solution to}$$

$$8x^3 - 6x - 1 = 0.$$

END OF SOLUTIONS!

Generally well done using various methods.

Generally well done.

Rarely done overall.

Students must use (i) & (ii) to achieve any marks.